

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

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Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-72098

Which of the following changes is the smallest?

- ☐ -8°C to -4°C
☐ -5°C to -2°C
☐ -2°C to 3°C
☐ 3°C to 7°C

2. ☒ Number and Algebra, NAP-42666

Which is the best estimate for $26 \times 14 + 77 - 32$?

- ☐ $20 \times 10 + 70 - 30$
☐ $20 \times 10 + 80 - 30$
☐ $30 \times 10 + 80 - 30$
☐ $30 \times 20 + 80 - 40$

3. ☒ Number and Algebra, NAP-13257

Freya and Eloise were playing a game where they both started with zero points.

Freya's final score was 320 points and Eloise's final score was -80 .

How many more points did Freya have than Eloise?

- -400 -240 240 400
☐ ☐ ☐ ☐

4. ☒ Number and Algebra, NAP-60272

1102 is a tri-square number because

$$1^2 + 1^2 + 0^2 = 2$$

The next tri-square number after 1102 is

- 1104 1113 1209 1126
☐ ☐ ☐ ☐

5. ☒ Number and Algebra, NAP-25119

$$6b - 19 = 3b + 8$$

What is the value of b in this equation?

$b =$

6. ☒ Number and Algebra, NAP-89767

Harriet solved the following equation:

$$7x - 2 = 4$$

Which of the following could be two lines of her solution?

- $7x = 2$ $7x = 2$ $7x = 6$ $7x = 6$
 $x = \frac{2}{7}$ $x = \frac{7}{2}$ $x = \frac{6}{7}$ $x = \frac{7}{6}$
☐ ☐ ☐ ☐

7. ☒ Number and Algebra, NAP-60333

Which value of A is closest to the missing number in this equation?

$$(A + 89.29) \div 5.12 = 48.4$$

- -80 80 130 160 250
☐ ☐ ☐ ☐ ☐

8. ☒ Number and Algebra, NAP-83853

Bobby wants to multiply 25×44 .

Which of these expressions will give Bobby the correct answer?

- ☐ $(20 \times 40) + (5 \times 4)$
- ☐ $(20 \times 4) + (40 \times 5)$
- ☐ $(20 \times 44) + (5 \times 44)$
- ☐ $(20 \times 40) + (20 \times 4) + (40 \times 5)$

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Worked Solutions

1. Number and Algebra, NAP-72098

Consider each option:

$$-8 \text{ to } -4 = 4^\circ \text{ change}$$

$$-5 \text{ to } -2 = 3^\circ \text{ change}$$

$$-2 \text{ to } 3 = 5^\circ \text{ change}$$

$$3 \text{ to } 7 = 4^\circ \text{ change}$$

$\therefore -5^\circ \text{ C to } -2^\circ \text{ C is the smallest change.}$

2. Number and Algebra, NAP-42666

Round 26 up to 30

$\Rightarrow$ round 14 down to 10

Round 77 up to 80

Round 32 down to 30

$\therefore$ Best estimate is $30 \times 10 + 80 - 30$

3. Number and Algebra, NAP-13257

$$\begin{aligned}\text{Difference} &= 320 - (-80) \\ &= 400\end{aligned}$$

4. Number and Algebra, NAP-60272

$$\begin{aligned}1^2 + 1^2 + 1^2 &= 1 + 1 + 1 \\ &= 3\end{aligned}$$

$\therefore 1113$ is the next tri-square number.

5. Number and Algebra, NAP-25119

$$\begin{aligned}6b - 19 &= 3b + 8 \\ 3b &= 27 \\ b &= 9\end{aligned}$$

6. Number and Algebra, NAP-89767

$$7x = 6$$

$$x = \frac{6}{7}$$

7. Number and Algebra, NAP-60333

$$(A + 89.29) \div 5.12 = 48.4$$

$$(A + 89.29) = 48.4 \times 5.12$$

$$\approx 50 \times 5 \text{ (using approximation)}$$

$$\approx 250$$

$$\therefore A \approx 250 - 89.29$$

$$\approx 250 - 90$$

$$\approx 160$$

8. Number and Algebra, NAP-83853

$$25 \times 44 = (20 \times 44) + (5 \times 44)$$